Teacher Wu Kuan | See It at a Glance #15 — tan θ Is Just Slope

English lecture notes · YouTube Short M2xZuWENROk

1. The trap question

What does tan θ actually mean?

2. The method

• Line through the origin at angle θ with the positive x-axis.
• Drop a vertical from any point: right triangle with rise and run.
• tan θ = opposite / adjacent = rise / run = slope.

3. Parent check method

Ask your child: “For a 45° line, how much do you go up for each step right?” (1 → tan 45° = 1.)

Practice

1. A line through the origin makes 45° with the positive x-axis. What is its slope?
2. A line through the origin makes 60° with the positive x-axis. What is its slope?
3. The line $y = (\sqrt{3}/3)x$ makes what angle with the positive x-axis?
4. The point (4, 3) is on a line through the origin at angle θ. Find tan θ.
5. A line through the origin has slope -1. What angle does it make with the positive x-axis?

Answers

1. 1
2. √3
3. 30°
4. 3/4
5. 135°

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